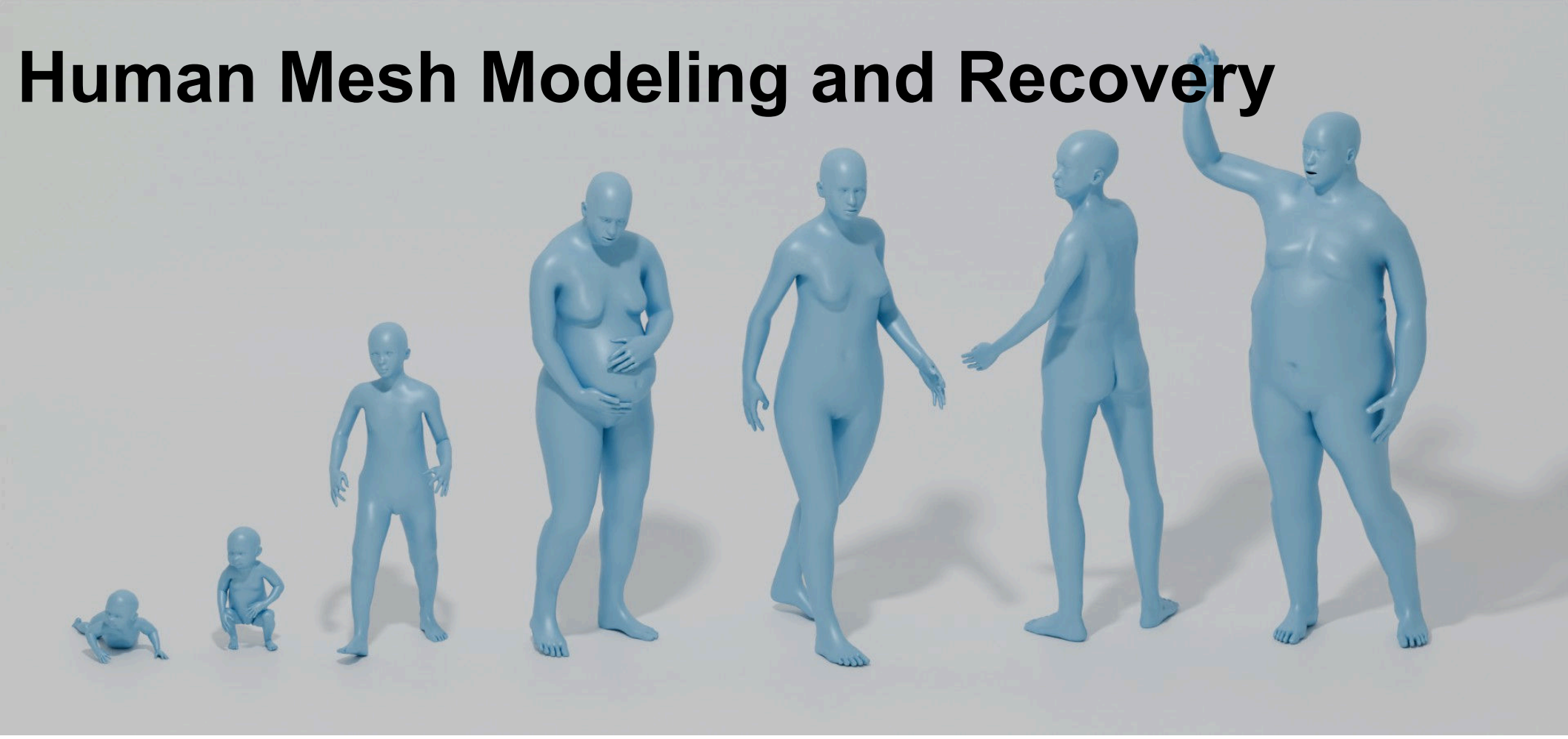
Human Mesh Modeling and Recovery



Grenoble Data Science Meetup
2026 Sept 24th

NAVER LABS Europe
Romain Brégier


NAVER
LABS

Dealing with humans





High Level Pipeline

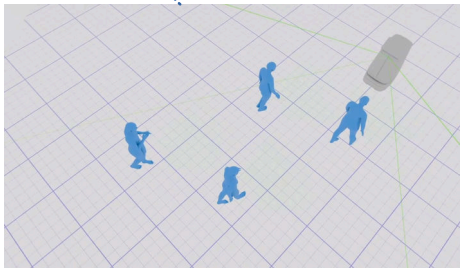


Sensor

System

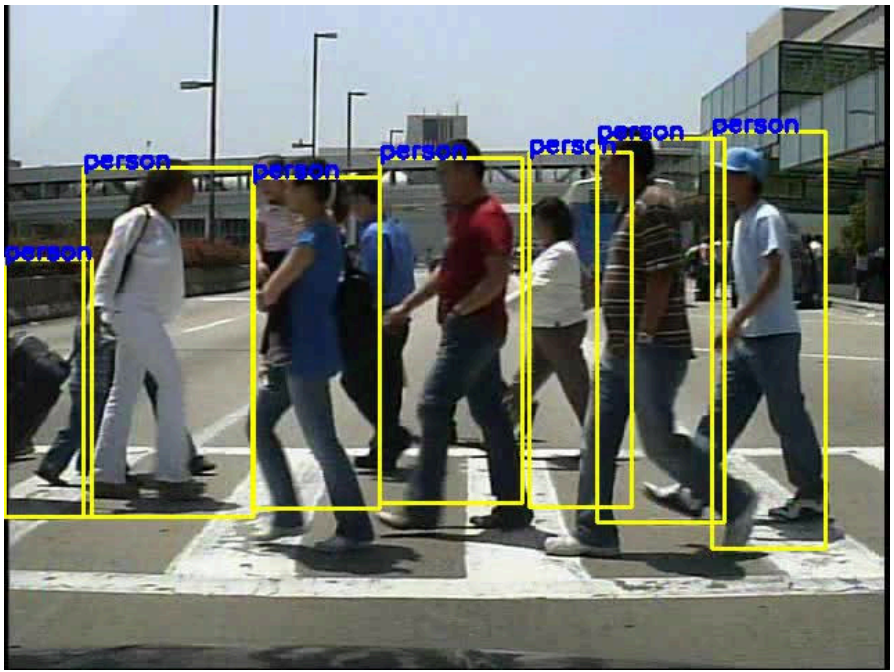
Action (turn / stop / ...)

Interpretability / alignment / harness



Model human pose, motions, actions, ...

Coarse human Model



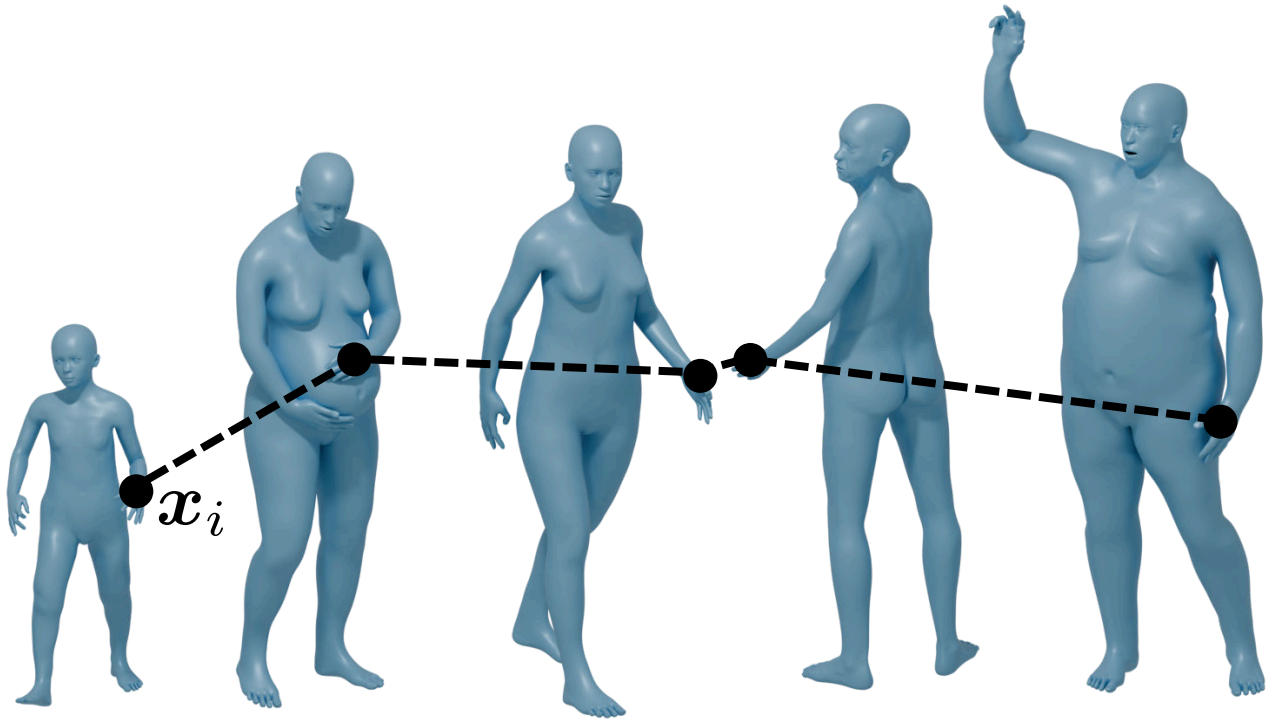
YOLO
(Redmon et al. CVPR2016)

Unstructured hi-res model





3D morphable body model



Body landmarks consistent across poses and identities

$$\boldsymbol{X} = (\boldsymbol{x}_i \in \mathbb{R}^3)_{i=1\ldots V}$$

$$\boldsymbol{X} = f(\boldsymbol{\beta}, \boldsymbol{\theta})$$

$\boldsymbol{\beta}$ Body shape
$\boldsymbol{\theta}$ Pose

SCAPE (Anguelov et al. 2005)

Deep roots



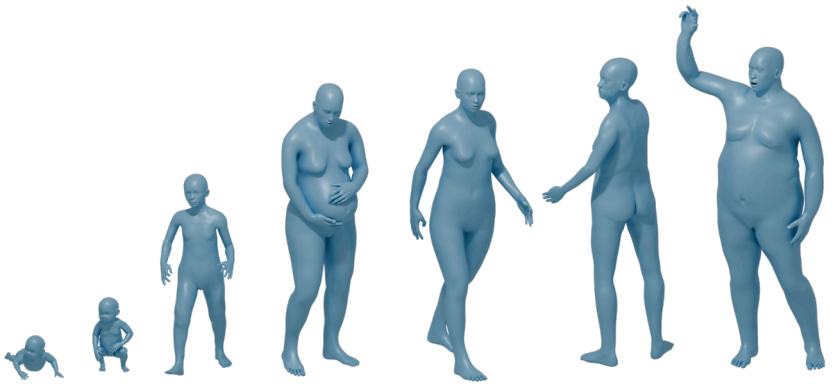


Hogg, David. “Model-Based Vision: A Program to See a Walking Person.” Image and Vision Computing 1, no. 1 (1983)

(Non-comprehensive) list

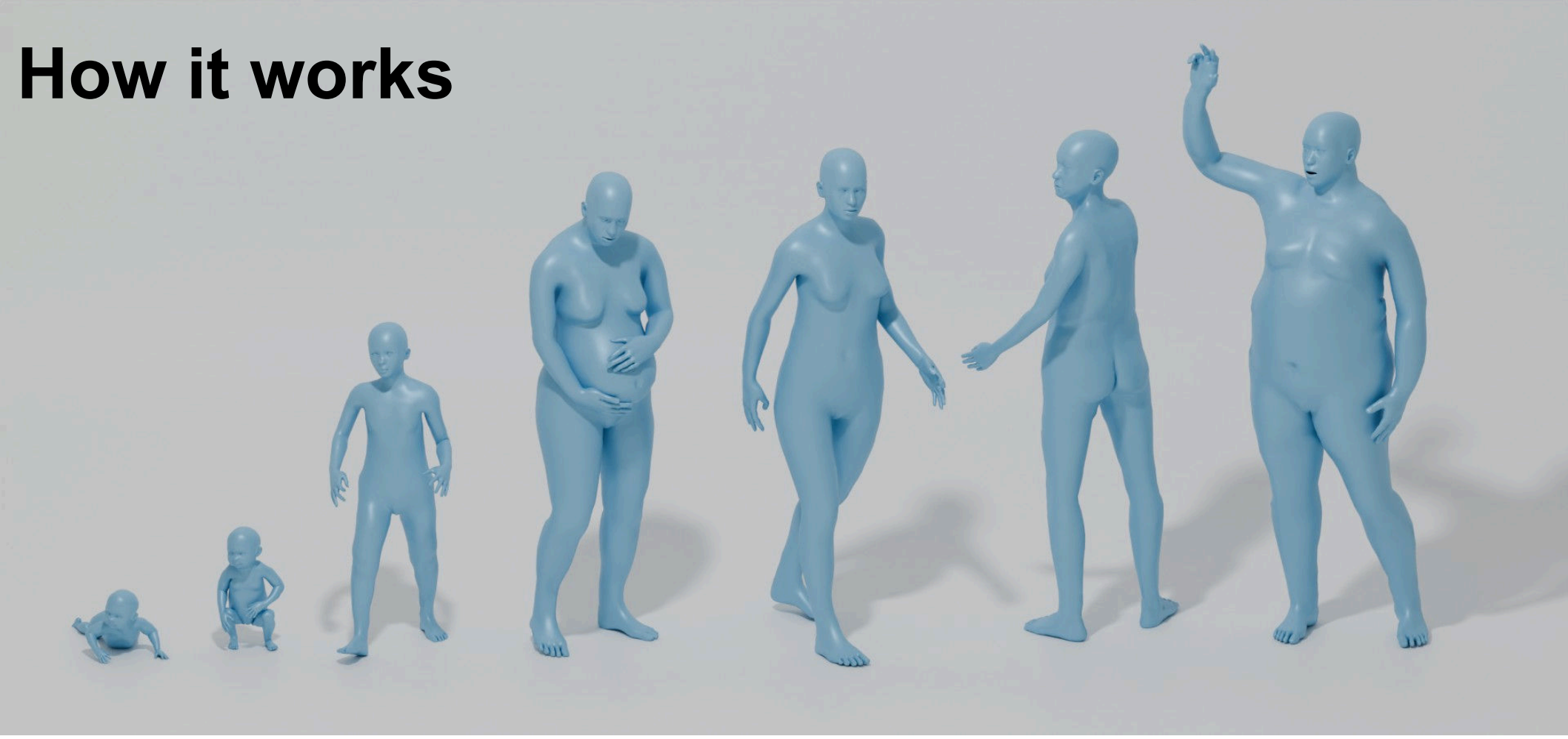


SMPL
(Loper et al. 2015)
Seminal work
Data driven
Adults-only

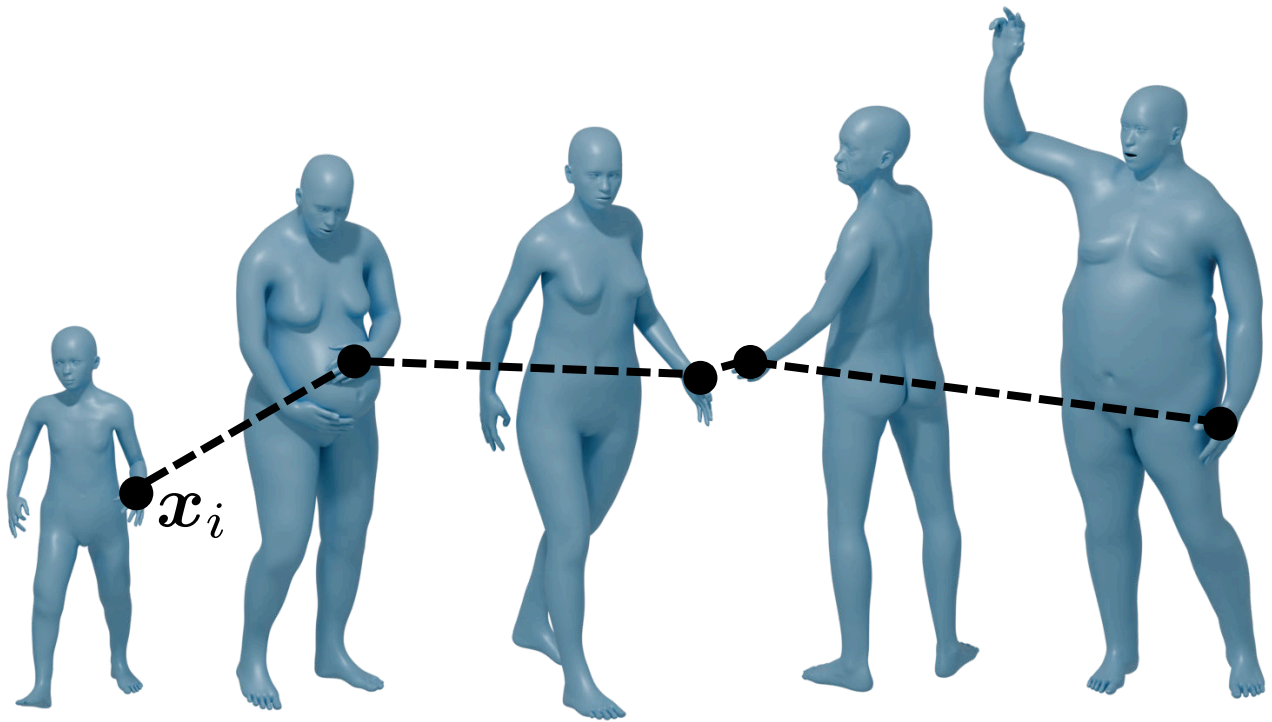


Anny
(Brégier et al. 2026)
Artists-based
All ages
Free (Apache 2.0)

How it works



3D Morphable Body model



Body landmarks consistent across poses and identities

$$\boldsymbol{X} = (\boldsymbol{x}_i \in \mathbb{R}^3)_{i=1\ldots V}$$

$$\boldsymbol{X} = f(\boldsymbol{\beta}, \boldsymbol{\theta})$$

$\boldsymbol{\beta}$ Body shape
$\boldsymbol{\theta}$ Pose

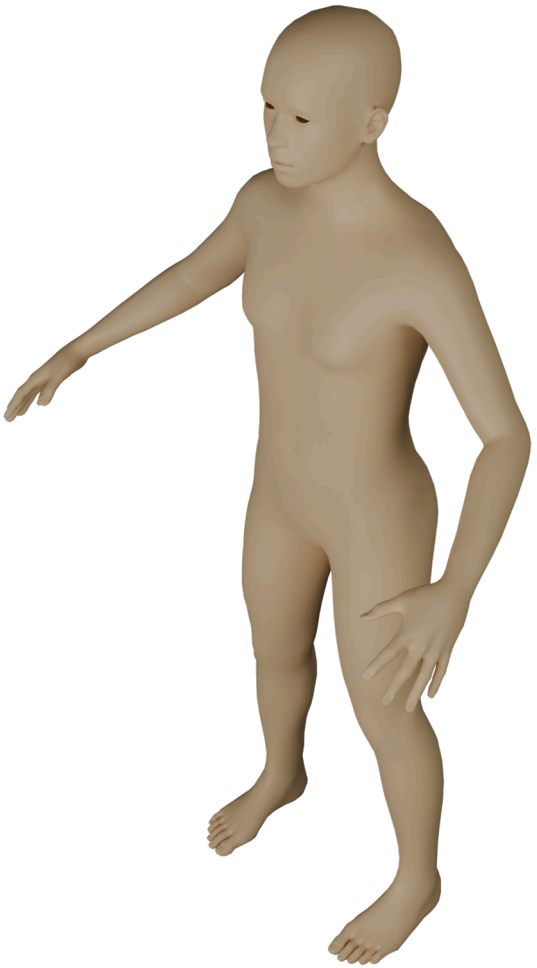
SCAPE (Anguelov et al. 2005)

Rigid body pose

$$\boldsymbol{x}_{\mathcal{W}} = \boldsymbol{T}_{body}\boldsymbol{x}_{body}$$

$$\boldsymbol{T}_{body} = \begin{pmatrix} \boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0}_{3\times 1} & 1 \end{pmatrix} \in SE(3)$$

$\boldsymbol{x}_{body}$ coordinates *wrt.* the *body*
$\boldsymbol{x}_{\mathcal{W}}$ absolute coordinates



Composed body



$\boldsymbol{x}_{\mathcal{W}} = \boldsymbol{T}_{\mathcal{B}} \boldsymbol{x}_{\mathcal{B}}$

$\boldsymbol{x}_{\mathcal{B}}$ coordinates *wrt.* the part $\mathcal{B}$

$\boldsymbol{T}_{\mathcal{B}}$ body part pose

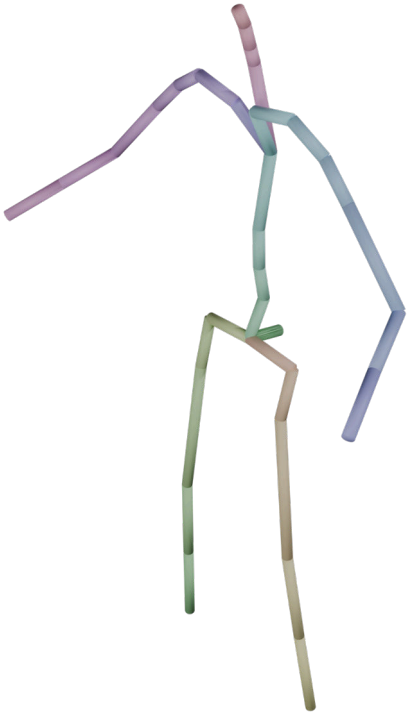
$\boldsymbol{x}_{\mathcal{W}}$ absolute coordinates

Skeletal animation

Forward kinematics:

$$\boldsymbol{T}_{bone} = \boldsymbol{T}_{parent}\boldsymbol{T}_{bone}^{local}$$

$$\boldsymbol{T}_{bone}^{local} = \begin{pmatrix} \boldsymbol{R} & \boldsymbol{0}_{3\times 1} \\ \boldsymbol{0}_{1\times 3} & 1 \end{pmatrix}$$

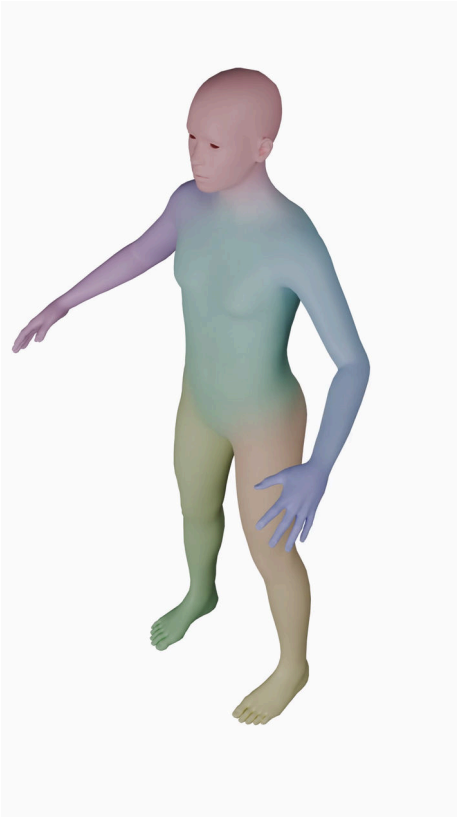




Linear Blend Skinning



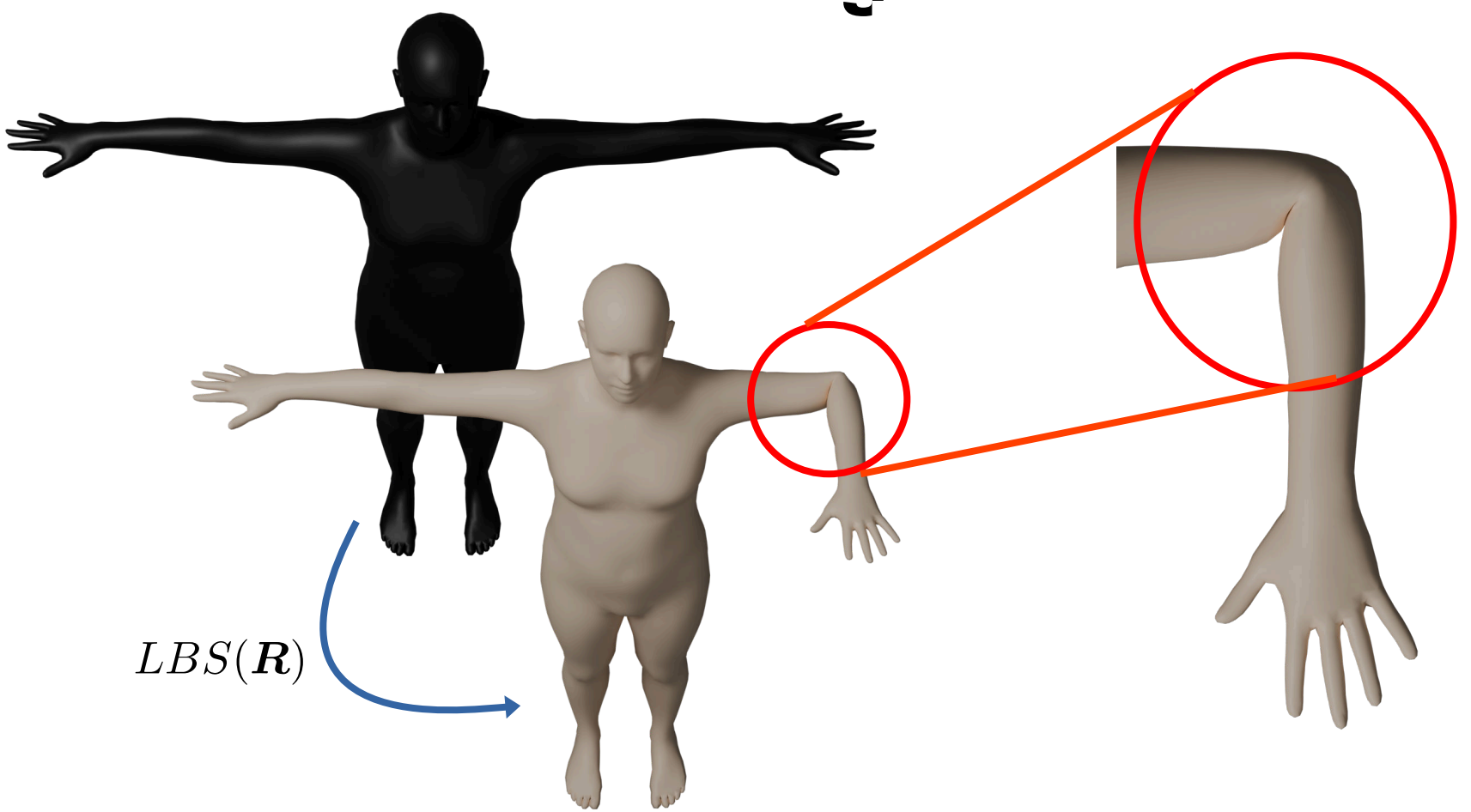
$$\boldsymbol{x}_{\mathcal{W}} = \boldsymbol{T}_{\mathcal{B}} \boldsymbol{x}_{\mathcal{B}}$$



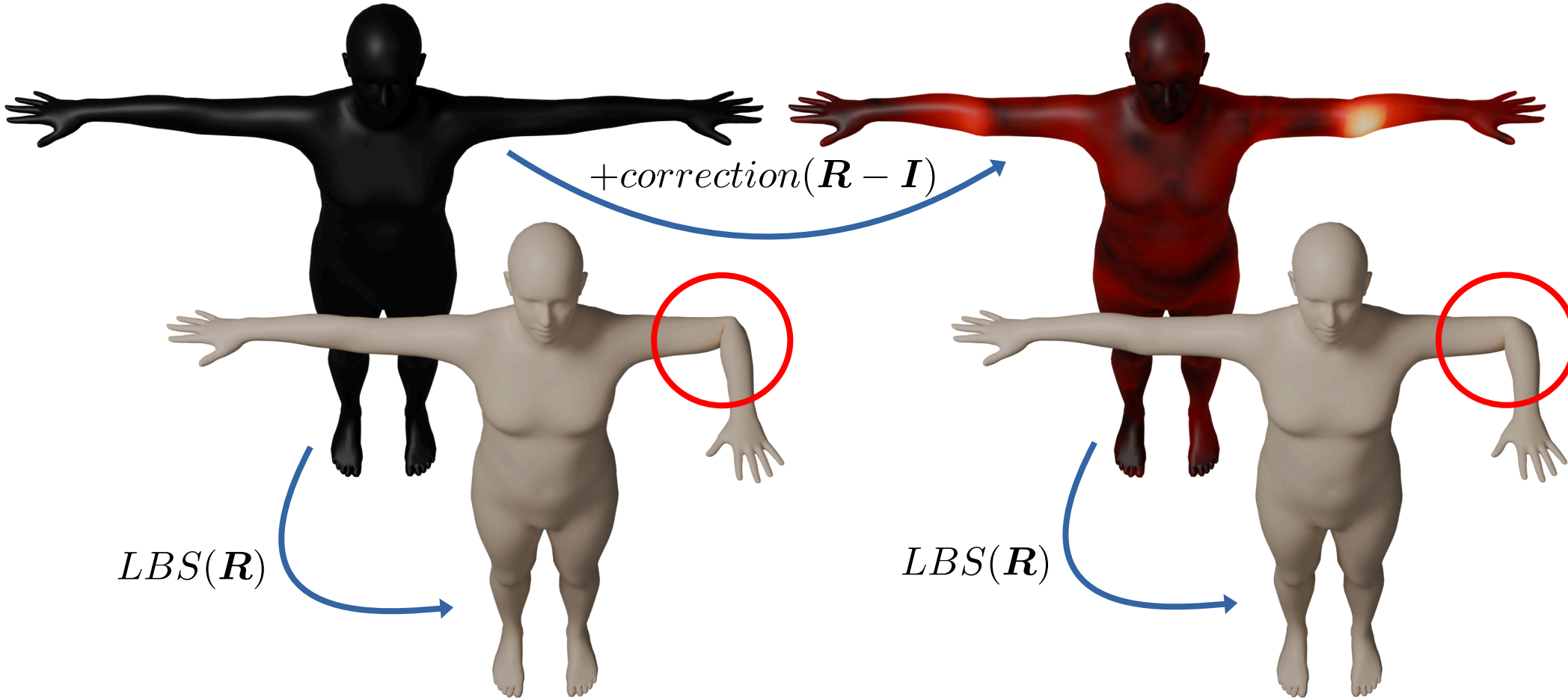
$$\boldsymbol{x}_{\mathcal{W}} = \sum_{\mathcal{B}} w_{\mathcal{B}} \boldsymbol{T}_{\mathcal{B}} \boldsymbol{x}_{\mathcal{B}}$$

$w_{\mathcal{B}}$ skinning weight $wrt.$ $\mathcal{B}$

Linear Blend skinning artifacts

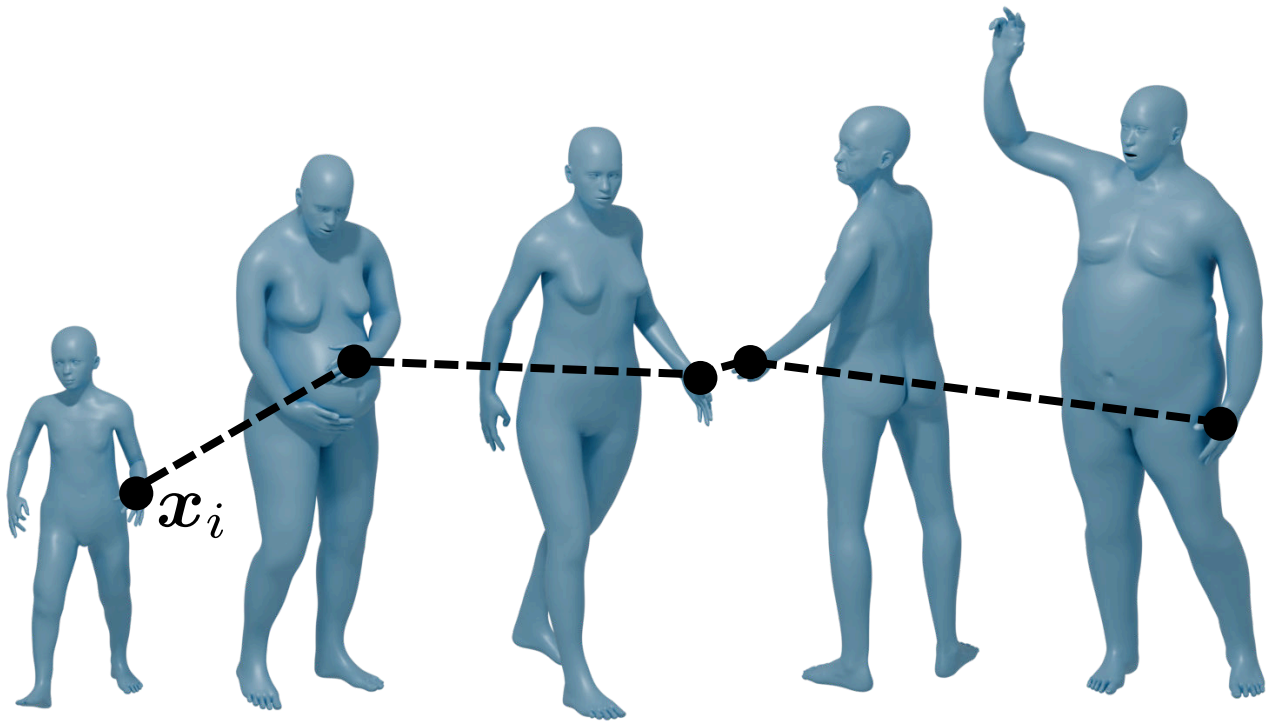


SMPL: Pose corrective deformations



SMPL (Loper et al. 2015)

Parametric surface model



Body landmarks consistent across poses and identities

$$\boldsymbol{X} = (\boldsymbol{x}_i \in \mathbb{R}^3)_{i=1\ldots V}$$

$$\boldsymbol{X} = f(\boldsymbol{\beta}, \boldsymbol{\theta})$$

$\boldsymbol{\beta}$ Body shape
$\boldsymbol{\theta}$ Pose

SCAPE (Anguelov et al. 2005)

Statistical shape modeling



Template mesh

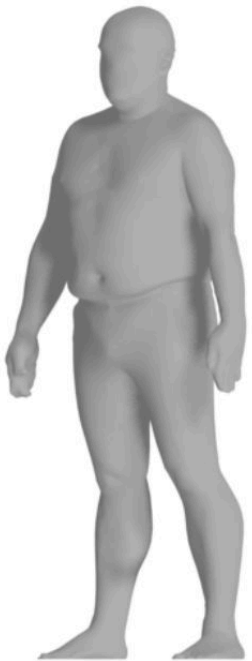
Fitting

Learned
parameterized
shape distribution

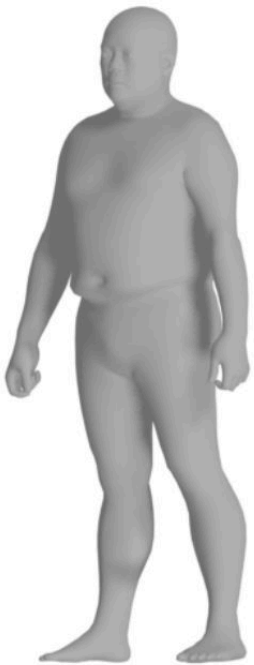


3D scans
(ATLAS)

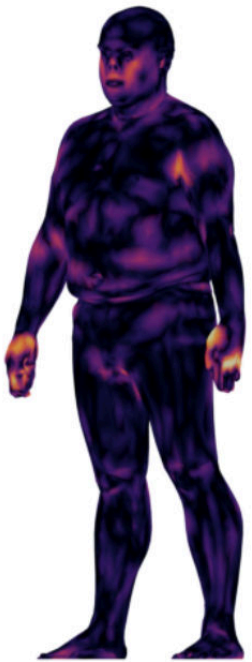
3D Scan Registration



Scan



Fit

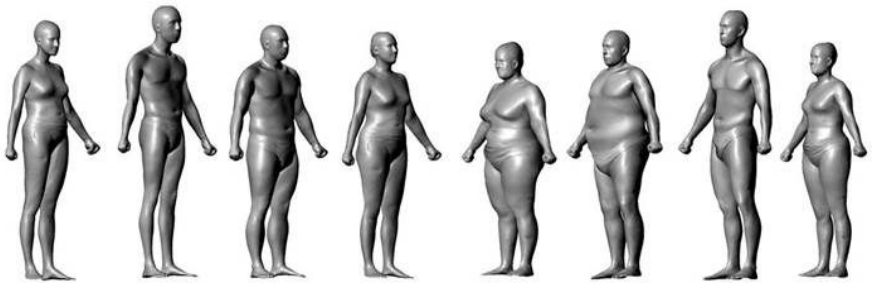


Fitting Error

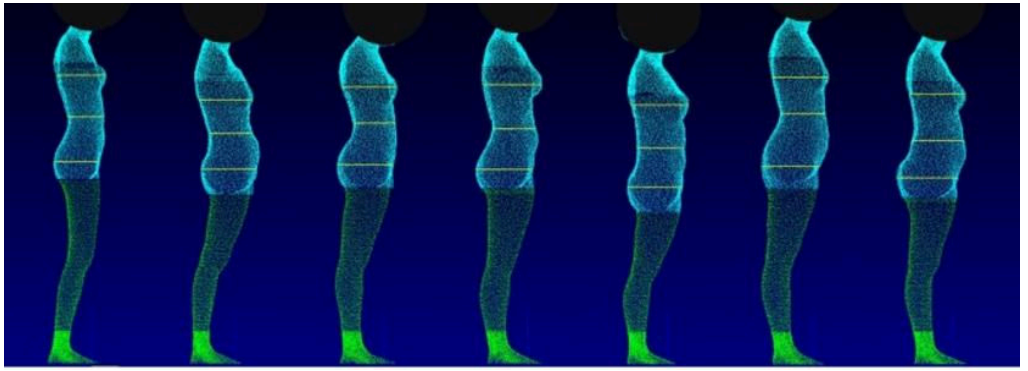


3DBodyTex dataset (Saint et al. 2018)

3D Body Scan datasets



CAESAR (~3.8k subjects)



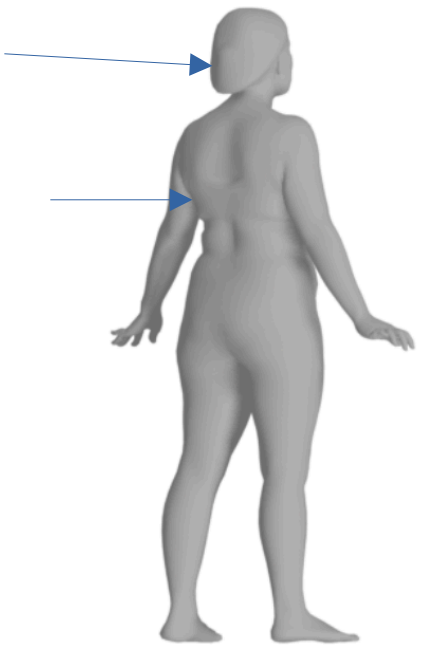
SizeUSA (~10k subjects)

+ other proprietary datasets

Scan-based shape modeling

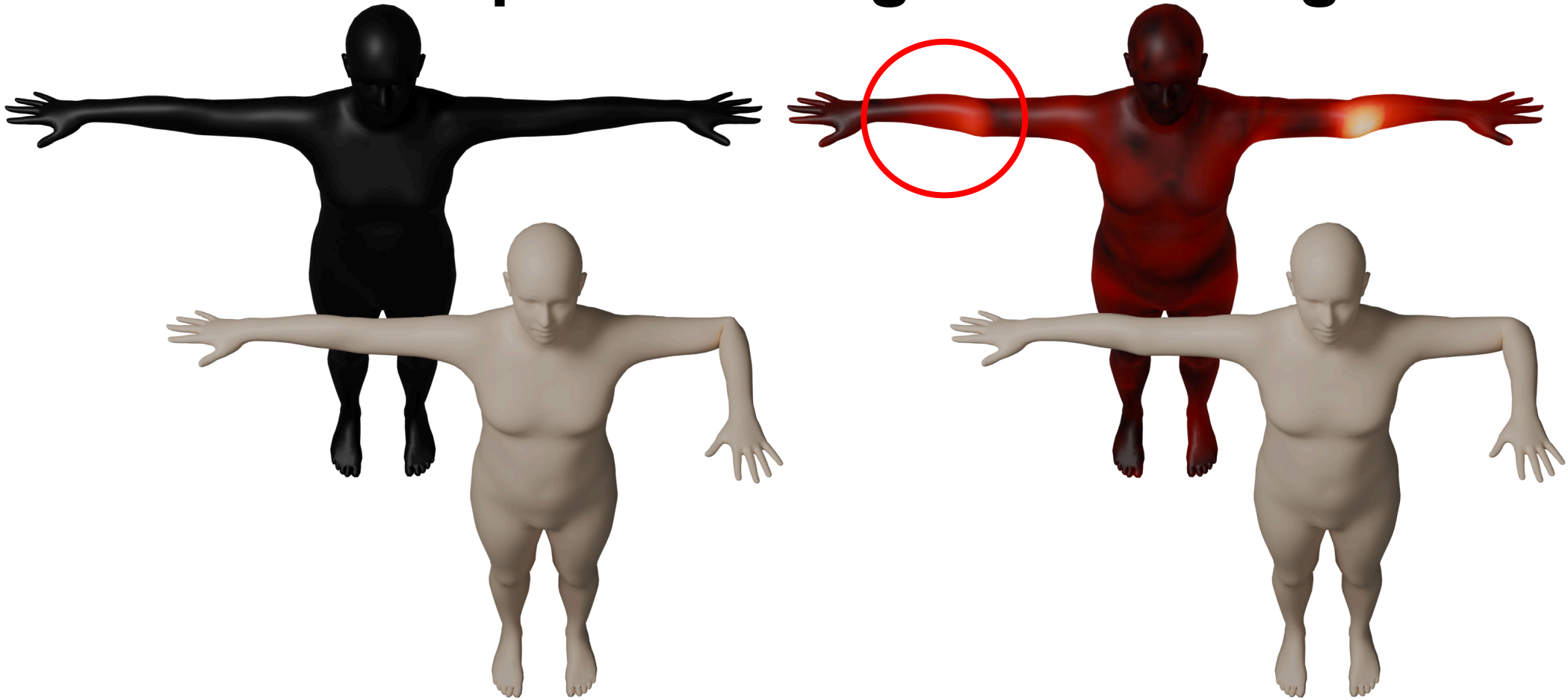
Pros:
- Great for in-distribution data

Cons:
- Out-of-distribution data
- Can over-fit

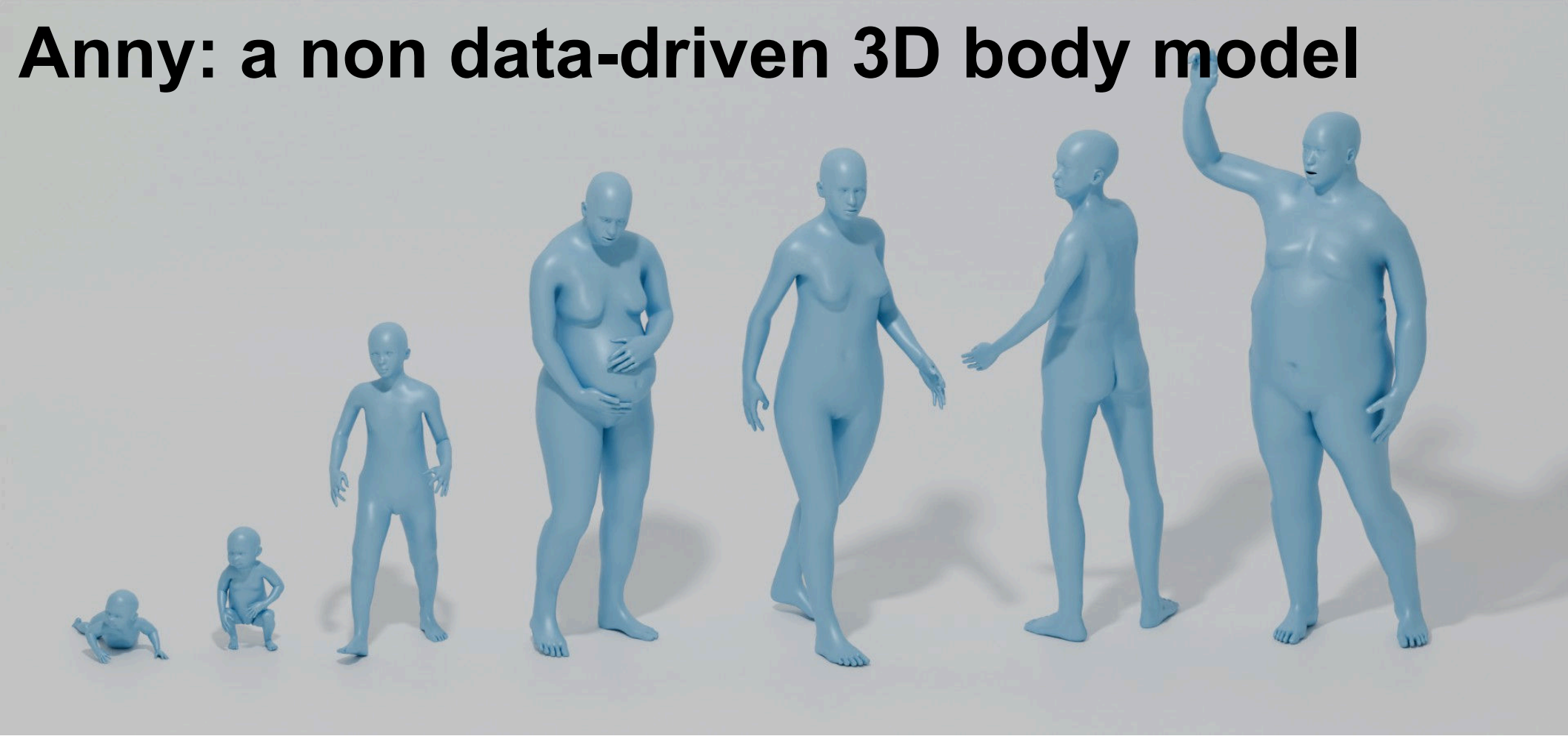


SMPL-X fit (300 shape components)

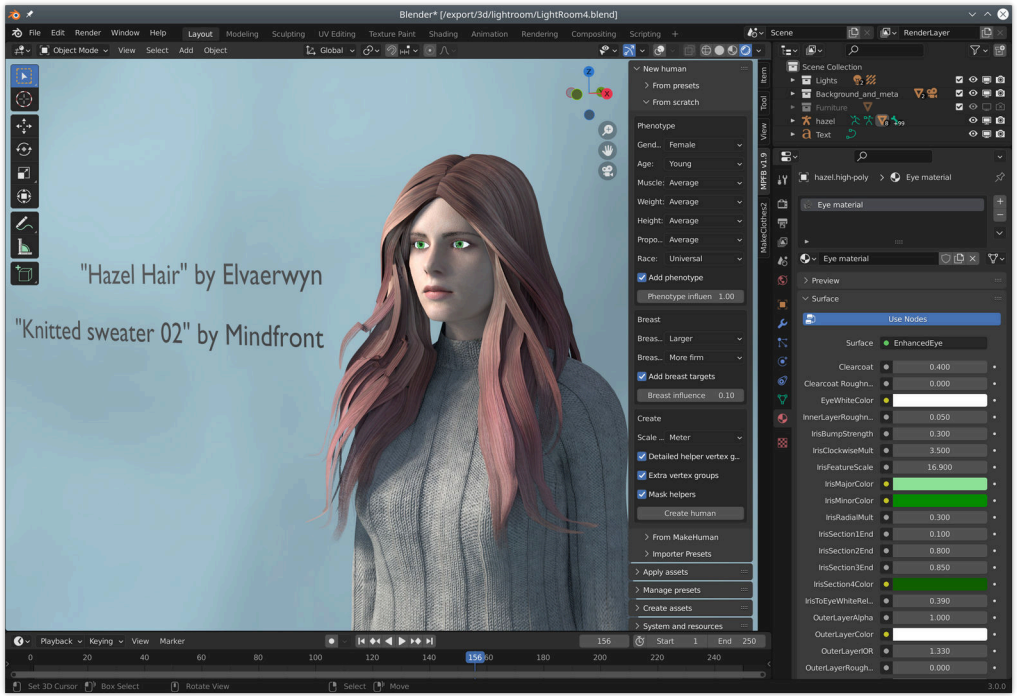
Statistical shape modeling “over-fitting”



Anny: a non data-driven 3D body model

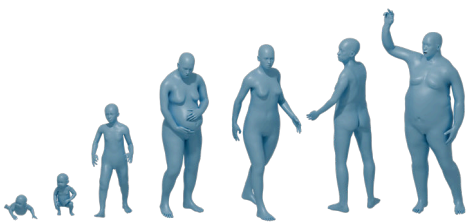


MakeHuman



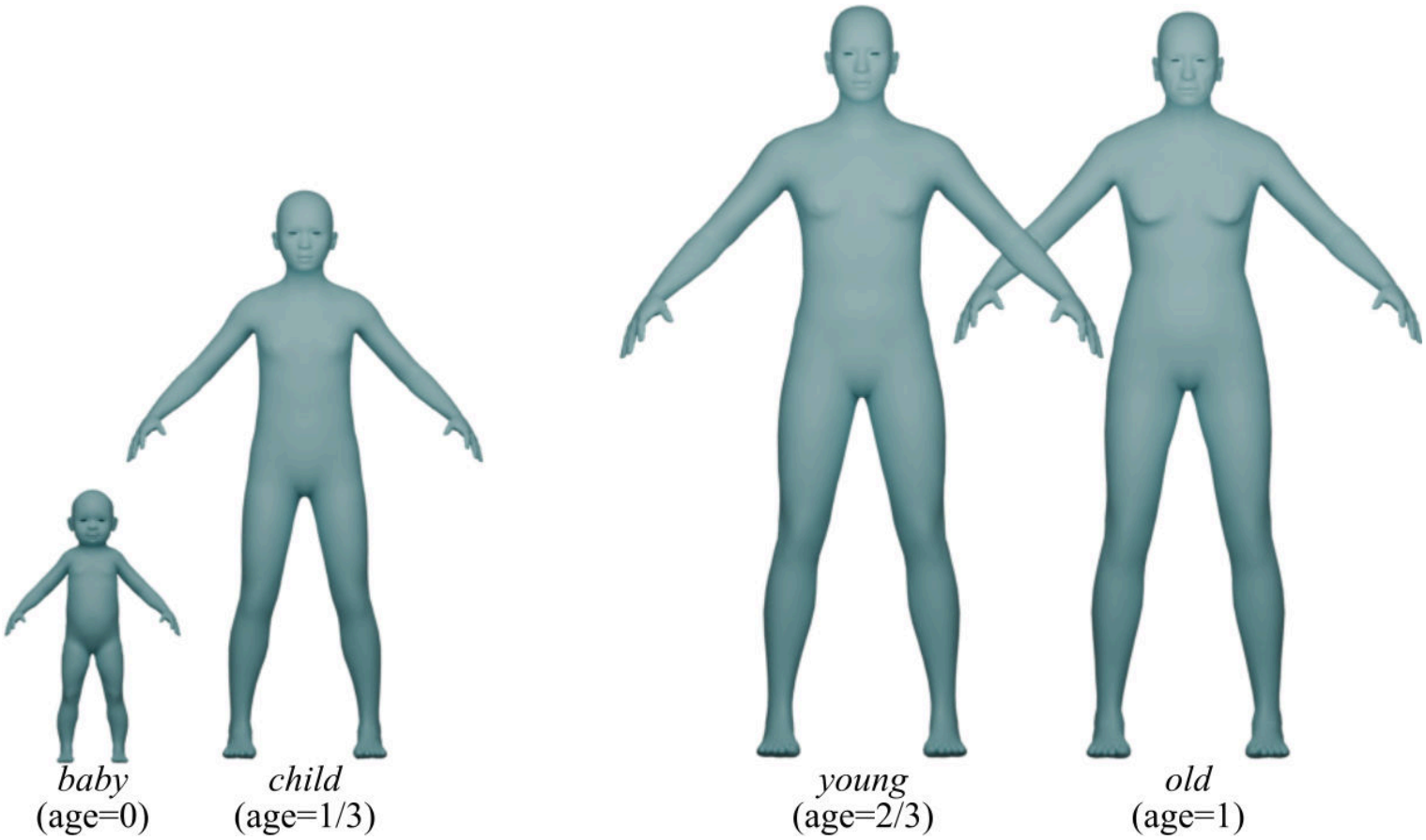
MPFB (2026)

\+ PyTorch

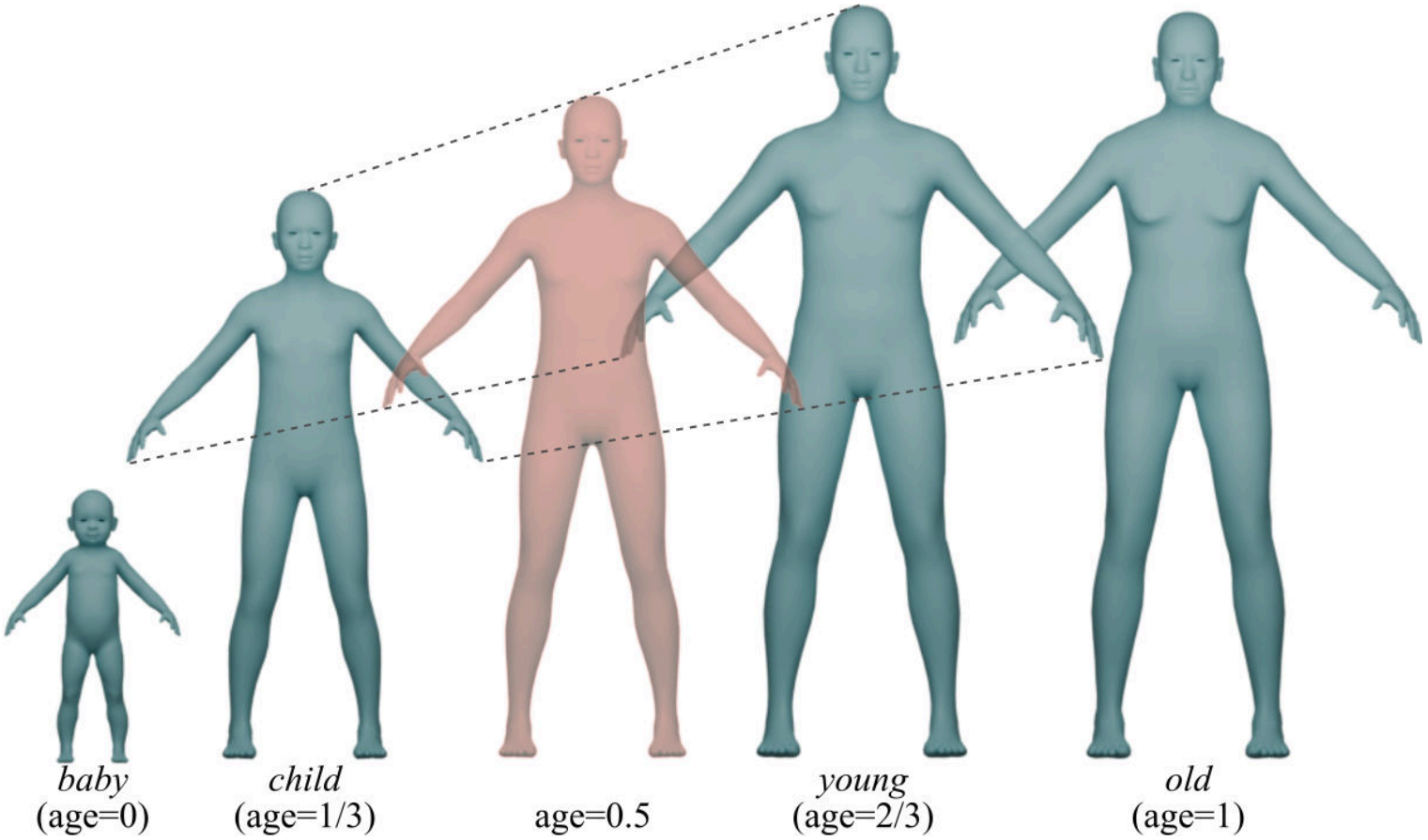


Anny

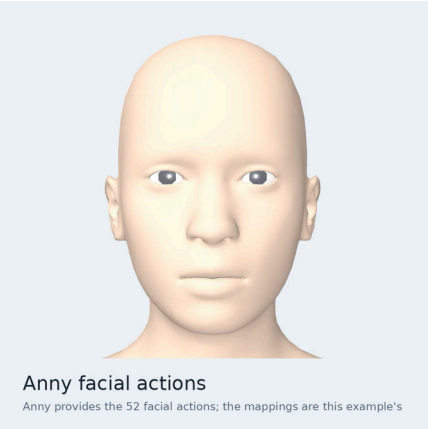
Prototypical shapes



Continuous shape space

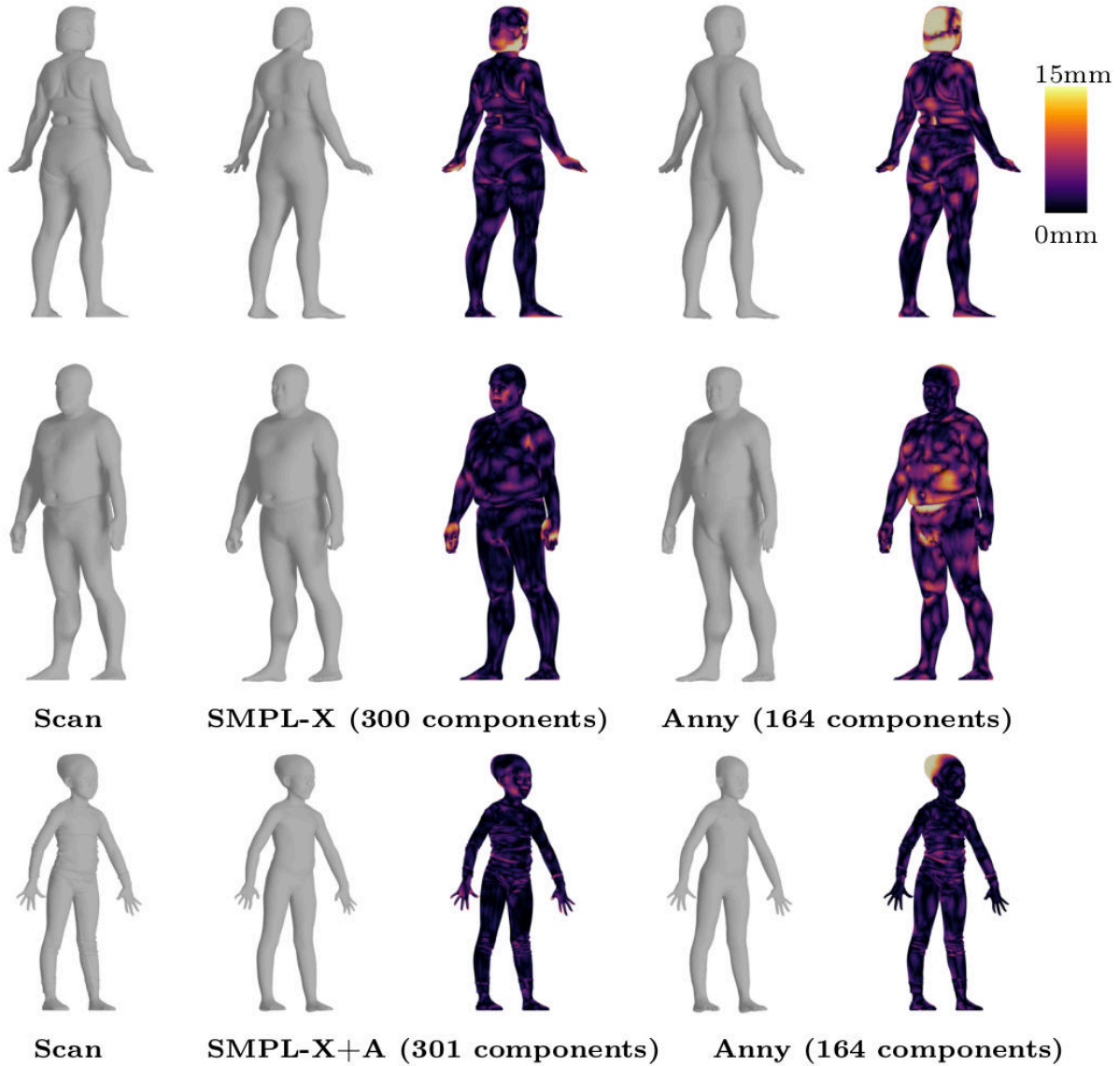


Continuous shape space

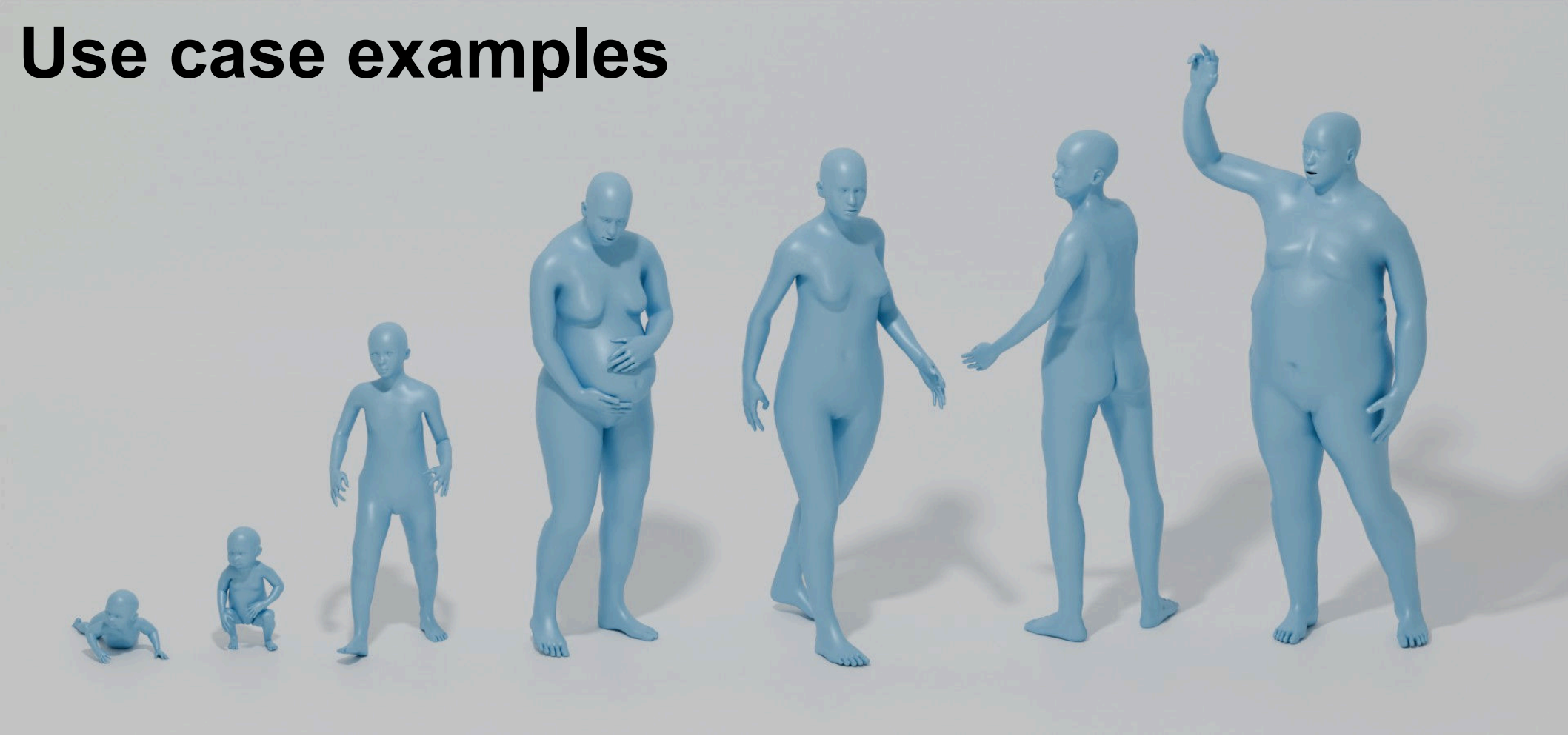




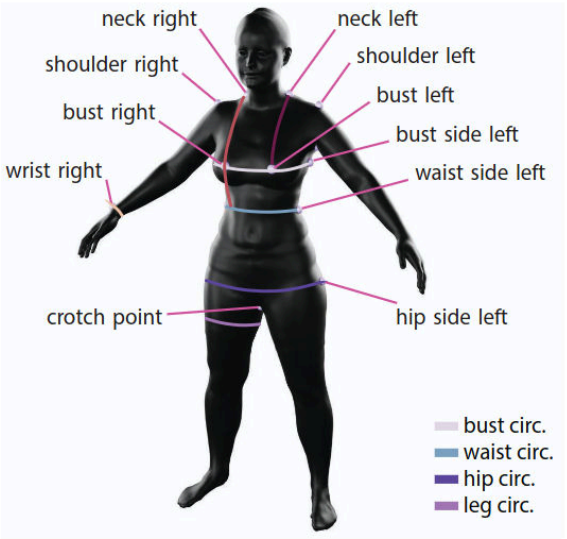
3D scan fitting



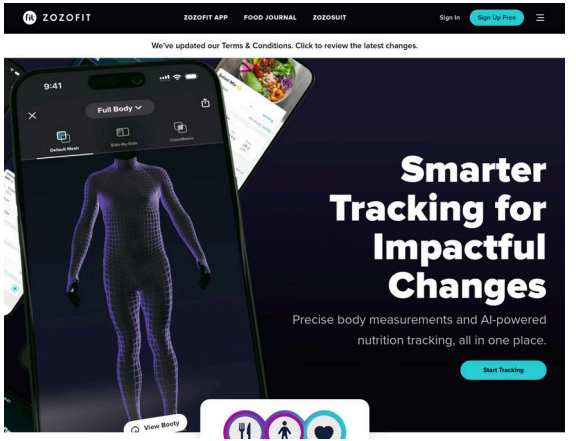
Use case examples



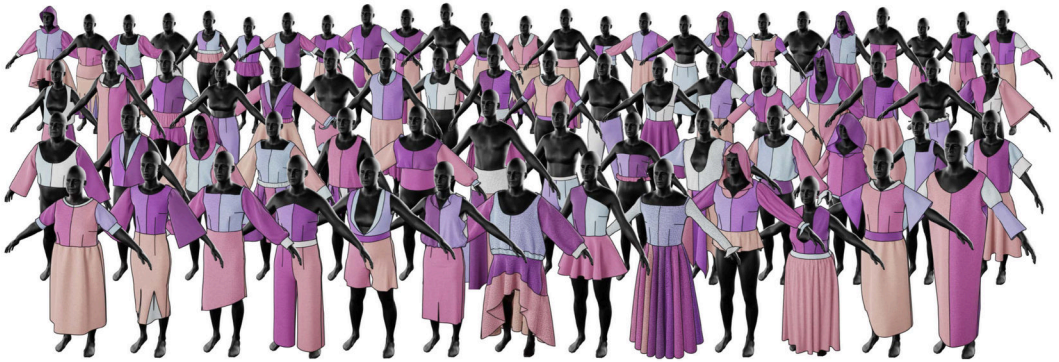
Body measurements





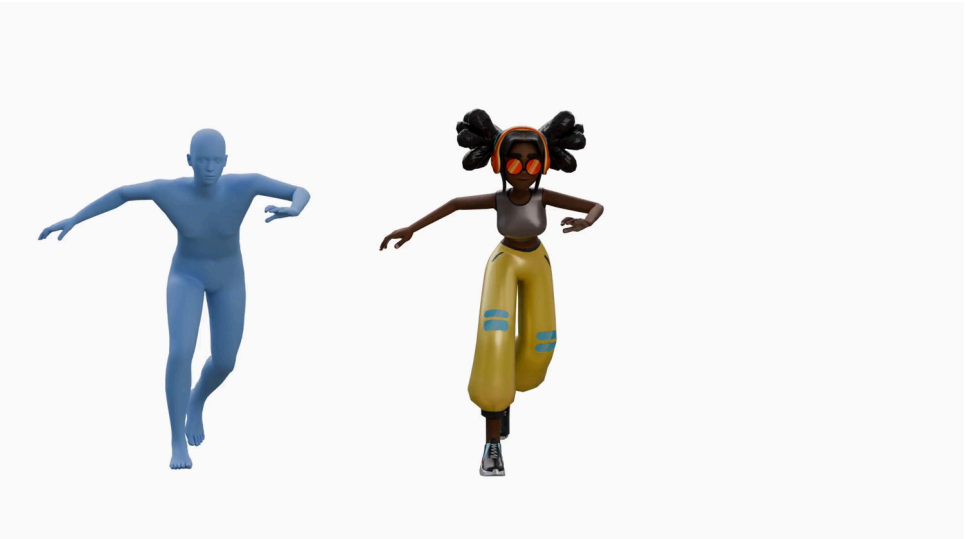


ZoZoFit

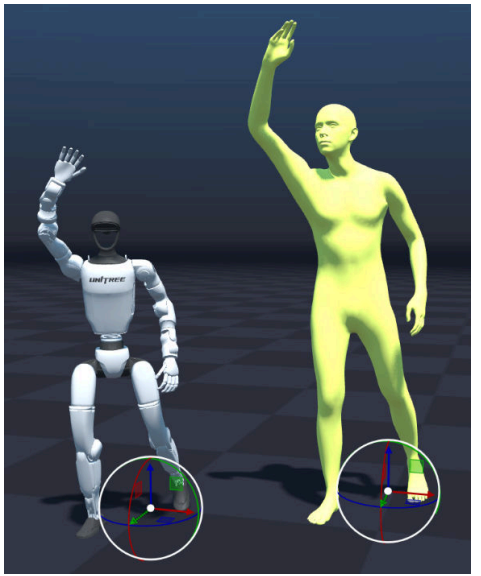


GarmentCodeData (Korosteleva et al. 2024)

Pose/motion transfer



Anny (Brégier et al. 2025)



SOMA retargeter (Nvidia 2026)

Biomechanics



OpenSim (Seth et al. 2018)

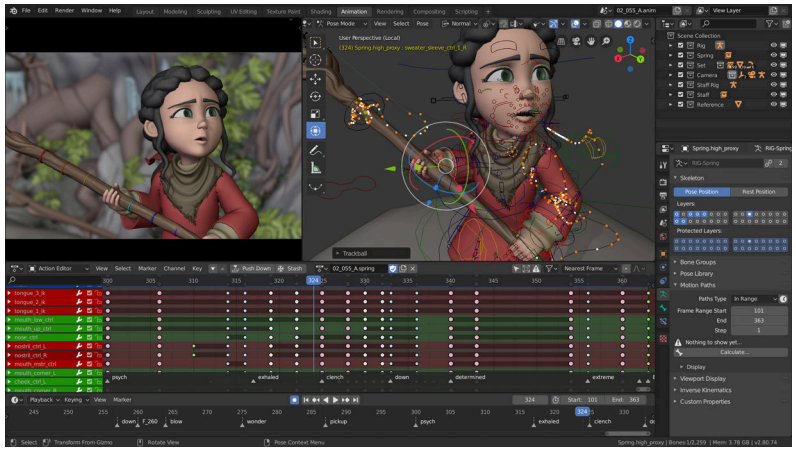


SKEL (Keller et al. 2023)



BOSS (Shetty et al. 2023)

User input

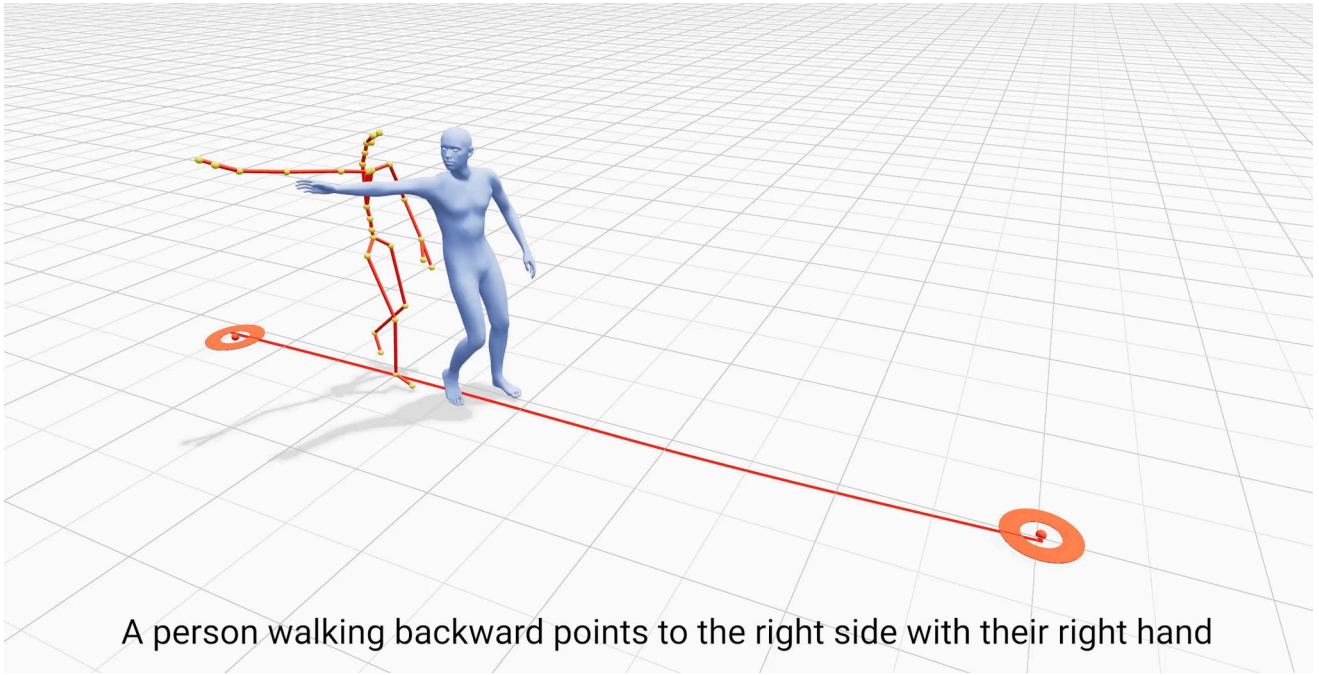


Blender (Spring 2019)



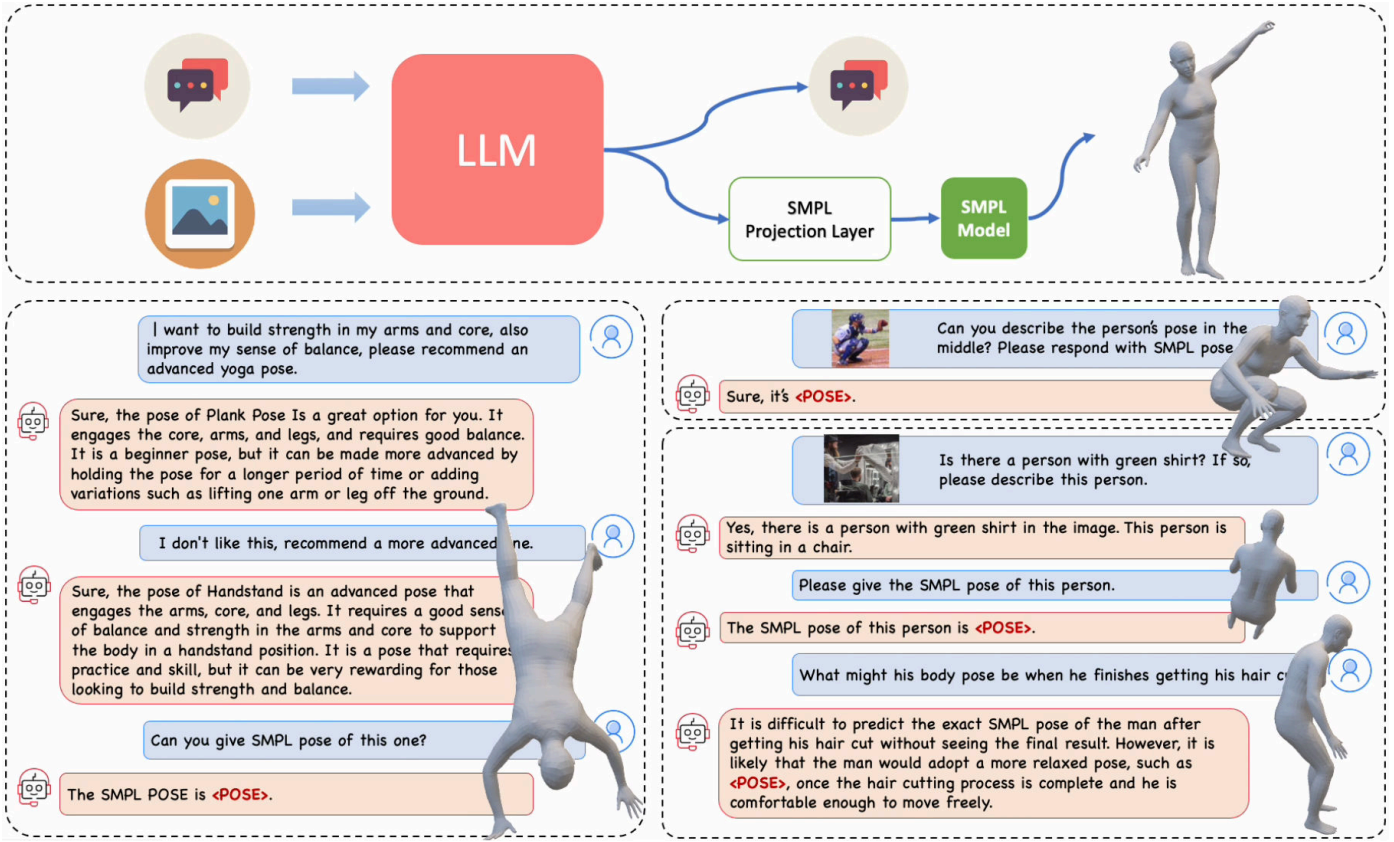
ControlNet (Zhang et al. ICCV23)

Motion generation



KIMODO (Rempe et al. 2026)

Description



ChatPose (Feng et al. CVPR2024)

Compress information







Naver MotionStage

Data generation



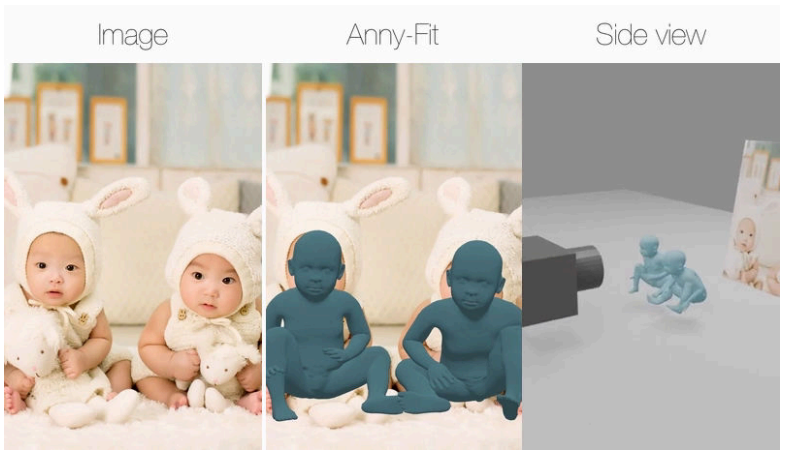
scenes

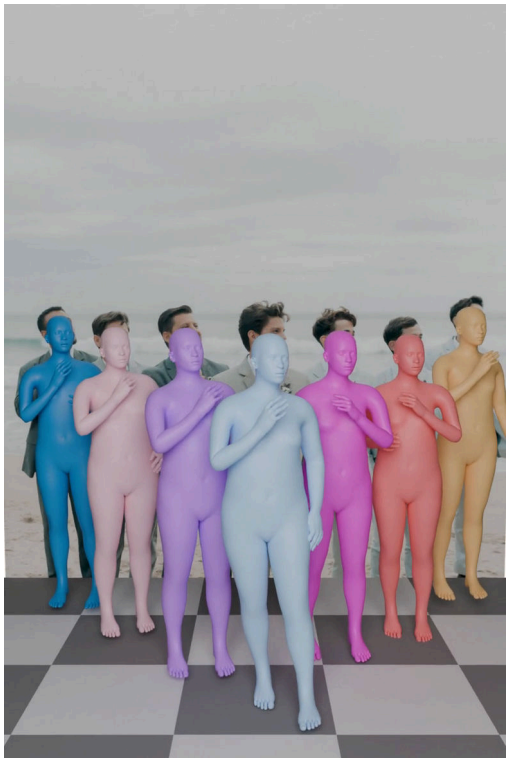
AMASS
poses

shapes

Human Mesh Recovery





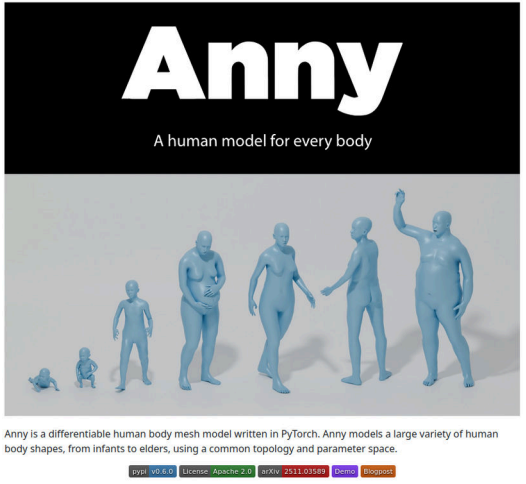


Fundamental limitation

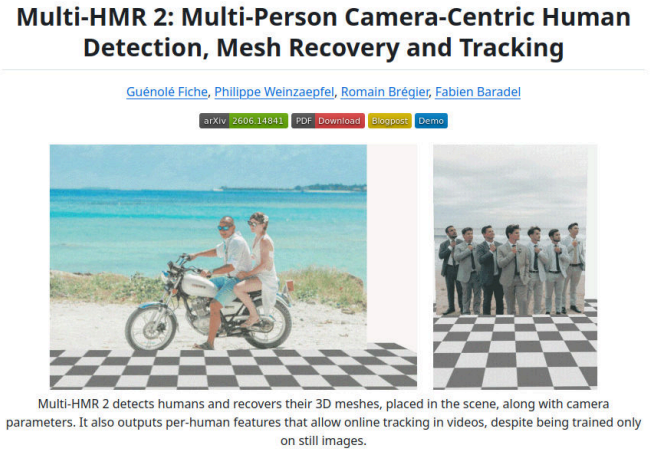




To go further



https://github.com/naver/anny



https://github.com/naver/multi-hmr2